

Task 1.

Use the Windows command prompt to run the command `ipconfig /all` on your computer and take a screenshot of the output. Identify and write down your IPv4 address, subnet mask, default gateway, and DNS servers as shown in the result.

```
C:\Users\sketan>ipconfig /all

Windows IP Configuration

Host Name . . . . . : LAPTOP-F1C8B05J
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No

Ethernet adapter Ethernet:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . : E3-WDS13.COM
Description . . . . . : Realtek PCIe GbE Family Controller
Physical Address. . . . . : BC-FC-E7-00-A5-F5
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Ethernet adapter Ethernet 2:

Connection-specific DNS Suffix . :
Description . . . . . : VirtualBox Host-Only Ethernet Adapter
Physical Address. . . . . : 0A-00-27-00-00-06
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::d853:f51b:3466:c8a6%6(Preferred)
IPv4 Address. . . . . : 192.168.56.1(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :
DHCPv6 IAID . . . . . : 755630119
DHCPv6 Client DUID. . . . . : 00-01-00-01-2E-F6-4D-3E-BC-FC-E7-00-A5-F5
NetBIOS over Tcpip. . . . . : Enabled

Wireless LAN adapter Local Area Connection* 1:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
Physical Address. . . . . : 62-FF-9E-4F-CE-82
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 2:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
Physical Address. . . . . : 66-FF-9E-4F-CE-82
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . :
Description . . . . . : Realtek 8852BE Wireless LAN WiFi 6 PCI-E NIC
Physical Address. . . . . : 60-FF-9E-4F-CE-82
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
IPv6 Address. . . . . : 2401:4900:191a:baf1:796a:3c24:d9c4:6b15(Preferred)
IPv6 Address. . . . . : 2001:db8:1:1::10(Duplicate)
Temporary IPv6 Address. . . . . : 2401:4900:191a:baf1:3c53:a3c5:492e:cf89(Preferred)
Link-local IPv6 Address . . . . . : fe80::6974:45a0:8874:a935%13(Preferred)
IPv4 Address. . . . . : 10.183.128.156(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : 04 August 2026 15:39:21
Lease Expires . . . . . : 04 August 2026 18:09:47
Default Gateway . . . . . : fe80::9409:ff:fe1d:7683%13
                             10.183.128.21
DHCP Server . . . . . : 10.183.128.21
DHCPv6 IAID . . . . . : 157351838
DHCPv6 Client DUID. . . . . : 00-01-00-01-2E-F6-4D-3E-BC-FC-E7-00-A5-F5
DNS Servers . . . . . : 10.183.128.21
                             2401:4900:191a:baf1::5e
NetBIOS over Tcpip. . . . . : Enabled
```

Parameter	Example Value
IPv4 Address	192.168.56.1
Subnet Mask	255.255.255.0
Default Gateway	
DNS Server	10.183.128.21

Task 2.

Simulate a network connectivity issue by disconnecting your Wi-Fi or Ethernet cable, then use the ping command to check connectivity to www.google.com. Reconnect and repeat the ping. Write a short note on the difference in results and what each status means.

```
C:\Users\sketan>ping www.google.com
Ping request could not find host www.google.com. Please check the name and try again.

C:\Users\sketan>
```

```
C:\Users\sketan>ping www.google.com
Ping request could not find host www.google.com. Please check the name and try again.

C:\Users\sketan>ping www.google.com

Pinging www.google.com [2001:4860:482c:7700::] with 32 bytes of data:
Reply from 2001:4860:482c:7700::: time=96ms
Reply from 2001:4860:482c:7700::: time=103ms
Reply from 2001:4860:482c:7700::: time=115ms
Reply from 2001:4860:482c:7700::: time=125ms

Ping statistics for 2001:4860:482c:7700:::
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 96ms, Maximum = 125ms, Average = 109ms
```

When disconnected: The ping failed because the computer had no network connection or could not reach the internet.

When reconnected: The ping succeeded, showing that the internet connection had been restored and the destination server was reachable.

Task 3.

Open Task Manager or Resource Monitor on your system and monitor network usage for 5 minutes while streaming a YouTube video. Record the peak network usage value and note which process used the most bandwidth during this time.

Observation

Example

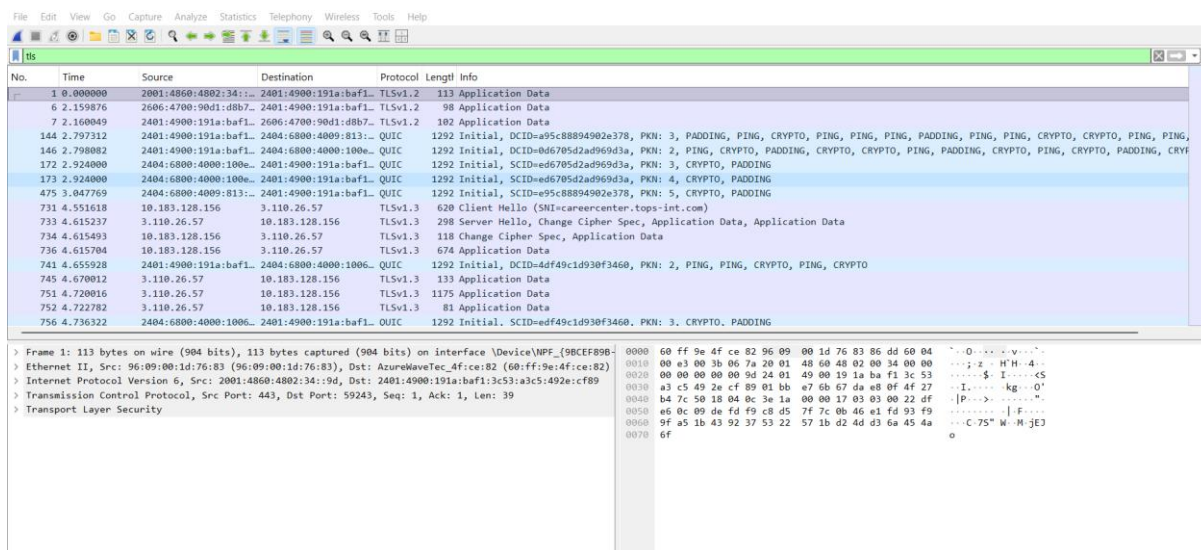
Peak Network Usage 5.1 Mbps

Process

Google Chrome
(chrome.exe)

Task 4.

Download and install Wireshark (or use an online packet capture demo if installation is not possible). Capture at least 30 seconds of network traffic while browsing Flipkart or Instagram Web, then filter the packets to show only HTTP or HTTPS traffic. List 3 types of information you can see in these filtered packets. Hint: Use the filter bar in Wireshark and try http or https as the filter keyword.



Types of Information

1. Source and destination IP addresses.
2. Protocol used (HTTP or TLS/HTTPS).

3. Packet size and timestamp.

Task 5.

Imagine your friend is complaining about slow internet while gaming online. Use ChatGPT or any AI tool to generate a troubleshooting checklist (at least 5 points) for optimizing home network speed and reliability. Paste the AI's response and highlight one suggestion you would try first.

Home Network Troubleshooting Checklist

Checklist

- Restart the modem and Wi-Fi router to clear temporary issues.
- Place the router in a central, open location away from walls and interference.
- Use a wired Ethernet connection instead of Wi-Fi for gaming whenever possible.
- Disconnect unnecessary devices using the network while gaming.
- Update the router firmware and network adapter drivers to the latest versions.
- Choose a less congested Wi-Fi channel or use the 5 GHz band if available.
- Close background downloads, cloud backups, and streaming applications before gaming.